Appendix F

Cross checks using 2D fits

With extensive help from Chanaka DeSilva with the fits and interpretation and from Anthony Timmins with interpretation, cross checks were done using 2D fits to ensure that the data were consistent if v2 were allowed to be a free parameter. The data were fit to a Gaussian in $\Delta\phi$ on the near-side (the *Ridge*), a Gaussian in $\Delta\phi$ and $\Delta\eta$ on the near-side (the jet-like correlation), a $\Delta\eta$ independent $\cos(\Delta\phi)$ term for the away-side, and a $\Delta\eta$-independent v_2 modulated background.

This was given by:

$$\frac{d^2N}{d\Delta\phi\Delta\eta} = B \quad \left(1 + 2\langle v_2^{trig}\rangle\langle v_2^{assoc}\rangle \cos 2\Delta\phi\right) + C\cos\Delta\phi \tag{F.1}$$

$$+ \quad \frac{Y_J}{2\sigma^J_{\Delta\phi}\sigma^J_{\Delta\eta}} e^{-\frac{\Delta\phi}{2\sigma^J_{\Delta\phi}}} e^{-\frac{\Delta\phi}{2\sigma^J_{\Delta\phi}}} \tag{F.2}$$

$$+ \quad \frac{dY_R}{d\Delta\eta}\frac{1}{\sqrt{2}\sigma^R_{\Delta\eta}} e^{-\frac{\Delta\phi}{2\sigma^J_{\Delta\phi}}}. \tag{F.3}$$

This is in contrast to the form assumed to describe the near-side in our analysis:

$$\frac{d^2N}{d\Delta\phi\Delta\eta} = B \quad \left(1 + 2\langle v_2^{trig}\rangle\langle v_2^{assoc}\rangle \cos 2\Delta\phi\right) \tag{F.4}$$

$$+ \quad \frac{Y_J}{2\sigma_{\Delta\phi}^J \sigma_{\Delta\eta}^J} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi}^J}} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi}^J}} \tag{F.5}$$

$$+ \quad \frac{dY_R}{d\Delta\eta} \frac{1}{\sqrt{2}\sigma_{\Delta\eta}^R} e^{-\frac{\Delta\phi}{2\sigma_{\Delta\phi}^J}}. \tag{F.6}$$

We do not consider the away-side in our analysis.

The fits were done in four different centrality bins in $Au + Au$ collisions at $\sqrt{s_{NN}}$ = 200 GeV. The jet-like yield is consistent with the method used in this paper. The widths in $\Delta\phi$ and $\Delta\eta$ are somewhat smaller in the fit method. v_2 is higher with the fit method for all bins except the 0-12% bin.

For comparison with the other method, the cos($\Delta\phi$) term is added to the ridge in order to get comparable terms. The cos($\Delta\phi$) term appears to lead to a higher *Ridge* yield. In the fit method, the cos($\Delta\phi$) term is considered to be an off-set from the v_2-modulated background, leading to a smaller background from the cos($2\Delta\phi$) term. This leads to a higher *Ridge* yield. In addition, the *Ridge* was genenerally fit to have a width of roughly 0.5 radians using the fit; this is likely because of the dip created by the cos($\Delta\phi$) term. Qualitatively the results are the same - the *Ridge* yield is larger in central collisions than peripheral collisions.

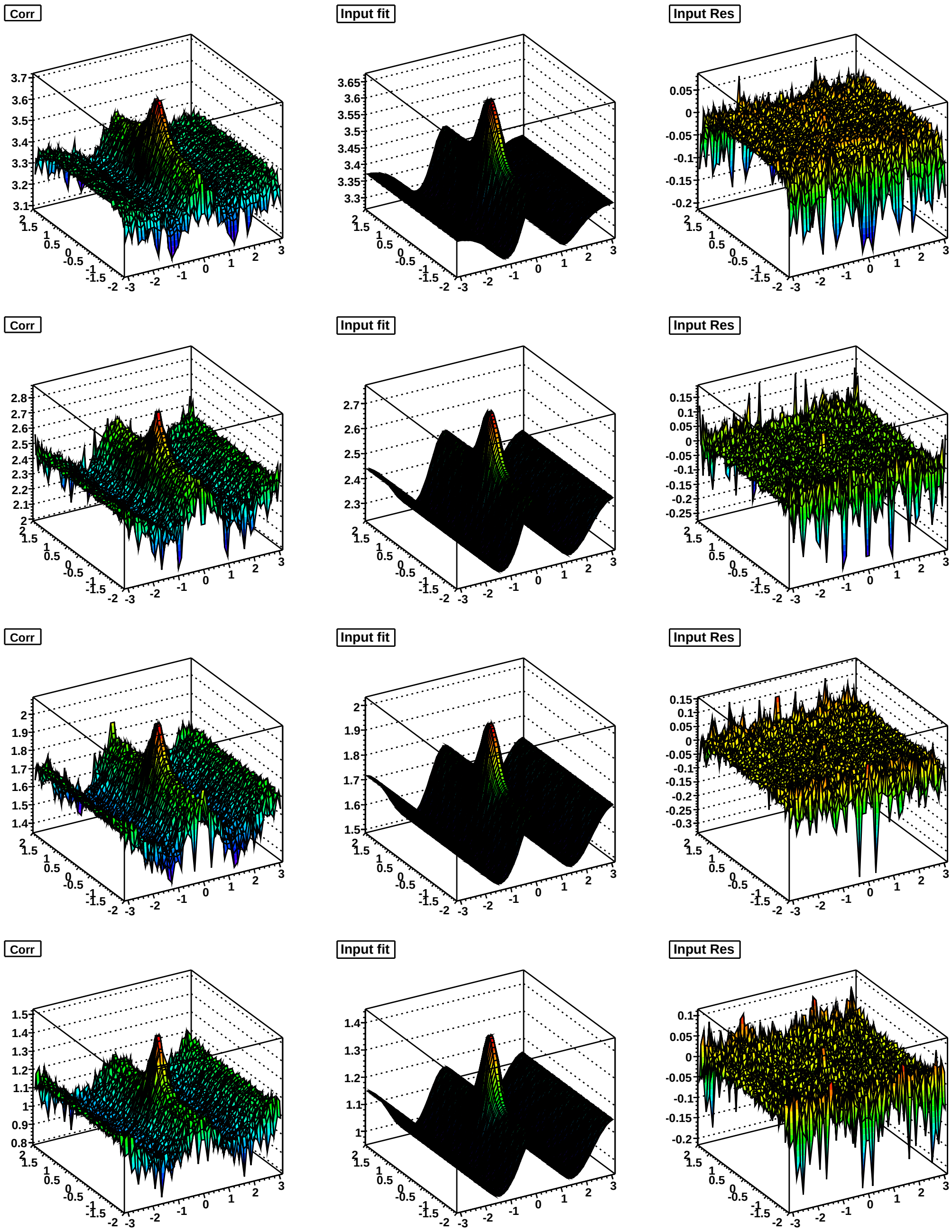

Figure F.1: 2D histogram (left), fit (middle), residuals (right) for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ and $3.0 < p_T^{trigger} < 6.0$ GeV/c in 0-12% (top), 10-20% (second down), 20-30% (third down), and 30-40% (bottom) $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV

0-12%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0045 (low), 0.0070 (avg.), 0.0100 (high)	-0.0138±0.0006
cos($\Delta\phi$) term coefficient	N/A	-0.097±0.001
near-side $\Delta\phi$ width	0.330 ± 0.006	0.318±0.002
near-side $\Delta\eta$ width	0.395 ± 0.008	0.363±0.003
jet-like yield	0.1298 ± 0.0015	0.134
Ridge yield	$0.270 \pm 0.006\ ^{+0.126}_{-0.160}$	0.664
10-20%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0148 (low), 0.0189 (avg.), 0.0236 (high)	0.0357±0.0014
cos($\Delta\phi$) term coefficient	N/A	-0.109±0.003
near-side $\Delta\phi$ width	0.304 ± 0.013	0.271±0.004
near-side $\Delta\eta$ width	0.328 ± 0.014	0.310±0.005
jet-like yield	0.1168 ± 0.0029	0.114
Ridge yield	$0.251 \pm 0.012\ ^{+0.167}_{-0.189}$	0.719
20-30%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0242 (low), 0.0294 (avg.), 0.0352 (high)	0.0589±0.0014
cos($\Delta\phi$) term coefficient	N/A	-0.088±0.003
near-side $\Delta\phi$ width	0.298 ± 0.011	0.273±0.003
near-side $\Delta\eta$ width	0.321 ± 0.012	0.296±0.004
jet-like yield	0.1226 ± 0.0027	0.117
Ridge yield	$0.191 \pm 0.011\ ^{+0.149}_{-0.166}$	0.507
30-40%		
	ZYAM	fit
v_2 (trig)*v_2 (assoc)	0.0278 (low), 0.0343 (avg.), 0.0414 (high)	0.0606±0.0012
cos($\Delta\phi$) term coefficient	N/A	-0.056±0.003
near-side $\Delta\phi$ width	0.287 ± 0.012	0.260±0.003
near-side $\Delta\eta$ width	0.293 ± 0.012	0.261±0.003
jet-like yield	0.1128 ± 0.0026	0.105
Ridge yield	$0.187 \pm 0.011\ ^{+0.119}_{-0.135}$	0.248

Table F.1: Fit parameters for 0-12% central $Au+Au$ at $\sqrt{s_{NN}} = 200$ GeV using both ZYAM and a fit

Appendix G

$p_T^{trigger}$ dependence for different collision energies

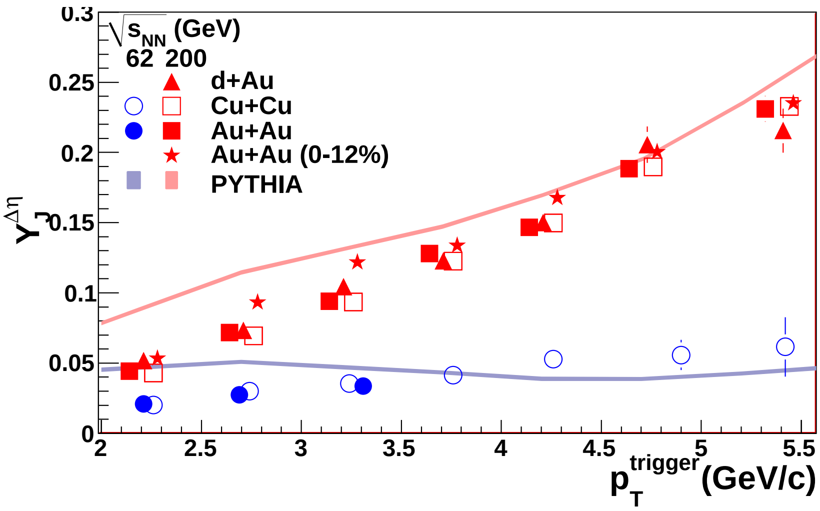

Figure G.1: $p_T^{trigger}$ dependence of Y_{Jet} for unidentified hadrons for 1.5 GeV/c $< p_T^{associated} < p_T^{trigger}$ for 0-60% central $Cu + Cu$ and 0-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 62$ GeV and minimum bias $d + Au$, 0-60% central $Cu + Cu$ and 40-80% central $Au + Au$ collisions at $\sqrt{s_{NN}} = 200$ GeV.

All points are done using bin counting and the $\Delta\eta$ method except for the three lowest pt bins for central $Au + Au$ at $\sqrt{s_{NN}} = 200$ GeV.

For the lowest p_T central $Au + Au$ $\sqrt{s_{NN}} = 200$ GeV data points, a fit in $\Delta\eta$ is